

Probability and Measure

Much of the theory of statistical inference can be appreciated without a detailed understanding of probability or measure theory. This book does not treat these topics with rigor. But some basic knowledge of them is quite useful. Much of the literature in statistics uses measure theory and is inaccessible to anyone unfamiliar with the basic notation. Also, the notation of measure theory allows one to merge results for discrete and continuous random variables. In addition, the notation can handle interesting and important applications involving censoring or truncation in which a random variable of interest is neither discrete nor continuous. Finally, the language of measure theory is necessary for stating many results correctly. In the sequel, measure-theoretic details are generally downplayed or ignored in proofs, but the presentation is detailed enough that anyone with a good background in probability should be able to fill in any missing details.

In this chapter measure theory and probability are introduced, and several of the most useful results are stated without proof.

1.1 Measures

A measure μ on a set $\mathcal{X}$ assigns a nonnegative value $\mu(A)$ to many subsets A of $\mathcal{X}$. Here are two examples.

Example 1.1. If $\mathcal{X}$ is countable, let

$$\mu(A) = \#A = \text{number of points in } A.$$

This μ is called *counting measure* on $\mathcal{X}$.

Example 1.2. Let $\mathcal{X} = \mathbb{R}^n$ and define

$$\mu(A) = \int \cdots \int_A dx_1 \cdots dx_n.$$

With $n = 1, 2$, or 3 , $\mu(A)$ is called the length, area, or volume of A , respectively. In general, this measure μ is called *Lebesgue measure* on $\mathbb{R}^n$. Actually, for some sets A it may not be clear how one should evaluate the integral “defining” $\mu(A)$, and, as we show, the theory of measure is fundamentally linked to basic questions about integration.

The measures in these examples differ from one another in an interesting way. Counting measure assigns mass to individual points, $\mu(\{x\}) = 1$ for $x \in \mathcal{X}$, but the Lebesgue measure of any isolated point is zero, $\mu(\{x\}) = 0$. In general, if $\mu(\{x\}) > 0$, then x is called an *atom* of the measure with mass $\mu(\{x\}) > 0$.

It is often impossible to assign measures to all subsets A of $\mathcal{X}$. Instead, the domain¹ of a measure μ will be a σ -field.

Definition 1.3. A collection $\mathcal{A}$ of subsets of a set $\mathcal{X}$ is a σ -field (or σ -algebra) if

1. $\mathcal{X} \in \mathcal{A}$ and $\emptyset \in \mathcal{A}$.
2. If $A \in \mathcal{A}$, then $A^c = \mathcal{X} - A \in \mathcal{A}$.
3. If $A_1, A_2, \dots \in \mathcal{A}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$.

The following definition gives the basic properties that must be satisfied for a set function μ to be called a measure. These properties should be intuitive for Examples 1.1 and 1.2.

Definition 1.4. A function μ on a σ -field $\mathcal{A}$ of $\mathcal{X}$ is a measure if

1. For every $A \in \mathcal{A}$, $0 \leq \mu(A) \leq \infty$; that is, $\mu : \mathcal{A} \rightarrow [0, \infty]$.
2. If $A_1, A_2, \dots$ are disjoint elements of $\mathcal{A}$ ($A_i \cap A_j = \emptyset$ for all $i \neq j$), then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

One interesting and useful consequence of the second part of this definition is that if measurable sets B_n , $n \geq 1$, are increasing ($B_1 \subset B_2 \subset \dots$), with union $B = \bigcup_{n=1}^{\infty} B_n$, called the limit of the sequence, then

$$\mu(B) = \lim_{n \rightarrow \infty} \mu(B_n). \quad (1.1)$$

This can be viewed as a continuity property of measures.

For notation, if $\mathcal{A}$ is a σ -field of subsets of $\mathcal{X}$, the pair $(\mathcal{X}, \mathcal{A})$ is called a *measurable space*, and if μ is a measure on $\mathcal{A}$, the triple $(\mathcal{X}, \mathcal{A}, \mu)$ is called a *measure space*.

A measure μ is *finite* if $\mu(\mathcal{X}) < \infty$ and σ -*finite* if there exist sets $A_1, A_2, \dots$ in $\mathcal{A}$ with $\mu(A_i) < \infty$ for all $i = 1, 2, \dots$ and $\bigcup_{i=1}^{\infty} A_i = \mathcal{X}$. All measures considered in this book are σ -finite.

¹ See Appendix A.1 for basic information and language about functions.

A measure μ is called a *probability measure* if $\mu(\mathcal{X}) = 1$, and then the triple $(\mathcal{X}, \mathcal{A}, \mu)$ is called a *probability space*. For probability (or other finite) measures, something analogous to (1.1) holds for decreasing sets. If measurable sets $B_1 \supset B_2 \supset \cdots$ have intersection $B = \bigcap_{n=1}^{\infty} B_n$, then

$$\mu(B) = \lim_{n \rightarrow \infty} \mu(B_n). \quad (1.2)$$

Example 1.1, continued. Counting measure given by $\mu(A) = \#A$ can be defined for any subset $A \subset \mathcal{X}$, so in this example, the σ -field $\mathcal{A}$ is the collection of all subsets of $\mathcal{X}$. This σ -field is called the *power set* of $\mathcal{X}$, denoted $\mathcal{A} = 2^{\mathcal{X}}$.

Example 1.2, continued. The Lebesgue measure of a set A can be defined, at least implicitly, for any set A in a σ -field $\mathcal{A}$ called the Borel sets of $\mathbb{R}^n$. Formally, $\mathcal{A}$ is the smallest σ -field that contains all “rectangles”

$$(a_1, b_1) \times \cdots \times (a_n, b_n) = \{x \in \mathbb{R}^n : a_i < x_i < b_i, i = 1, \dots, n\}.$$

Although there are many subsets of $\mathbb{R}^n$ that are not Borel, none of these sets can be written explicitly.

1.2 Integration

The goal of this section is to properly define integrals of “nice” functions f against a measure μ . The integral written as $\int f d\mu$ or as $\int f(x) d\mu(x)$ when the variable of integration is needed. To motivate later developments, let us begin by stating what integration is for counting and Lebesgue measure.

Example 1.5. If μ is counting measure on $\mathcal{X}$, then the integral of f against μ is

$$\int f d\mu = \sum_{x \in \mathcal{X}} f(x).$$

Example 1.6. If μ is Lebesgue measure on $\mathbb{R}^n$, then the integral of f against μ is

$$\int f d\mu = \int \cdots \int f(x_1, \dots, x_n) dx_1 \cdots dx_n.$$

It is convenient to view x as the vector $(x_1, \dots, x_n)'$ and write this integral against Lebesgue measure as $\int \cdots \int f(x) dx$ or $\int f(x) dx$.

The modern definition of integration given here is less constructive than the definition offered in most basic calculus courses. The construction is driven by basic properties that integrals should satisfy and proceeds arguing that for “nice” functions f these properties force a unique value for $\int f d\mu$. A key regularity property for the integrand is that it is “measurable” according to the following definition.

Definition 1.7. If $(\mathcal{X}, \mathcal{A})$ is a measurable space and f is a real-valued function on $\mathcal{X}$, then f is measurable if

$$f^{-1}(B) \stackrel{\text{def}}{=} \{x \in \mathcal{X} : f(x) \in B\} \in \mathcal{A}$$

for every Borel set B .

Although there are many functions that are not measurable, they cannot be stated explicitly. Continuous and piecewise continuous functions are measurable. A more interesting example is the function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = 1$ when x is an irrational number in $(0, 1)$, and $f(x) = 0$ otherwise. With the Riemann notion of integration used in basic calculus courses, for this function f , $\int f(x) dx$ is not defined. The more general methods presented here give the natural answer, $\int f(x) dx = 1$. In the sequel, functions of interest are generally presumed to be measurable.

The indicator function 1_A of a set A is defined as

$$1_A(x) = I\{x \in A\} = \begin{cases} 1, & x \in A; \\ 0, & x \notin A. \end{cases}$$

Here are the basic properties for integrals.

1. For any set A in $\mathcal{A}$, $\int 1_A d\mu = \mu(A)$.
2. If f and g are nonnegative measurable functions, and if a and b are positive constants,

$$\int (af + bg) d\mu = a \int f d\mu + b \int g d\mu. \quad (1.3)$$

3. If $f_1 \leq f_2 \leq \dots$ are nonnegative measurable functions, and if $f(x) = \lim_{n \rightarrow \infty} f_n(x)$, then

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

The first property provides the link between $\int f d\mu$ and the measure μ , the second property is linearity, and the third property is useful for taking limits of integrals.

Using the first two properties, if $a_1, \dots, a_m$ are positive constants, and if $A_1, \dots, A_m$ are sets in $\mathcal{A}$, then

$$\int \left(\sum_{i=1}^m a_i 1_{A_i} \right) d\mu = \sum_{i=1}^m a_i \mu(A_i).$$

Functions of this form are called *simple*. Figure 1.1 shows the graph of the simple function $1_{(1/2, \pi)} + 21_{(1, 2)}$. The following result asserts that nonnegative measurable functions can be approximated by simple functions.

Theorem 1.8. If f is nonnegative and measurable, then there exist nonnegative simple functions $f_1 \leq f_2 \leq \dots$ with $f = \lim_{n \rightarrow \infty} f_n$.

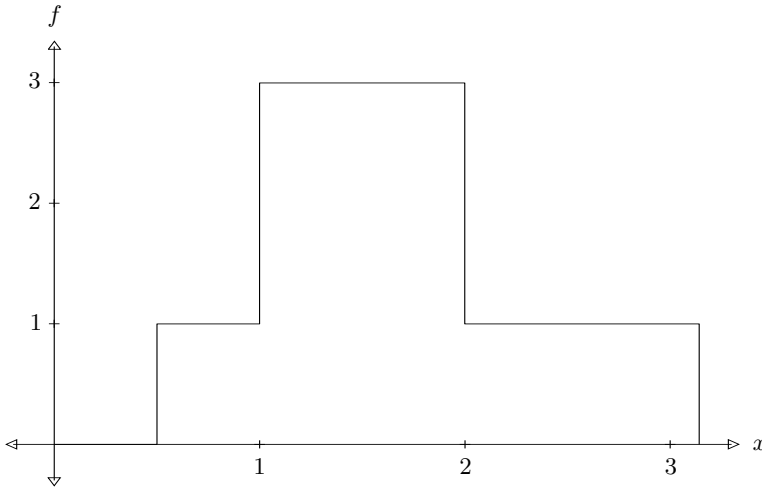


Fig. 1.1. The simple function $1_{(1/2, \pi)} + 21_{(1, 2)}$.

This result along with the third basic property for integrals allows us to integrate any nonnegative measurable function f , at least in principle. The answer is unique; different choices for the increasing sequence of simple functions give the same answer. To integrate a general measurable function f , introduce the positive and negative parts

$$f^+(x) = \max\{f(x), 0\} \quad \text{and} \quad f^-(x) = -\min\{f(x), 0\}.$$

Then f^+ and f^- are both nonnegative and measurable, and $f = f^+ - f^-$. The integral of f should generally be the difference between the integral of f^+ and the integral of f^- . This difference is ambiguous only when the integrals of f^+ and f^- are both infinite. So, if either $\int f^+ d\mu < \infty$ or $\int f^- d\mu < \infty$, we define

$$\int f d\mu = \int f^+ d\mu - \int f^- d\mu.$$

With this definition the linearity in (1.3) holds unless the right-hand side is formally $\infty - \infty$. Note also that because $|f| = f^+ + f^-$, this definition gives a finite value for $\int f d\mu$ if and only if

$$\int f^+ d\mu + \int f^- d\mu = \int |f| d\mu < \infty.$$

When $\int |f| d\mu < \infty$, f is called *integrable*.

1.3 Events, Probabilities, and Random Variables

Let P be a probability measure on a measurable space $(\mathcal{E}, \mathcal{B})$, so $(\mathcal{E}, \mathcal{B}, P)$ is a probability space. Sets $B \in \mathcal{B}$ are called *events*, points $e \in \mathcal{E}$ are called *outcomes*, and $P(B)$ is called the *probability* of B .

A measurable function $X : \mathcal{E} \rightarrow \mathbb{R}$ is called a *random variable*. The probability measure P_X defined by

$$P_X(A) = P(\{e \in \mathcal{E} : X(e) \in A\}) \stackrel{\text{def}}{=} P(X \in A)$$

for Borel sets A is called the *distribution* of X . The notation

$$X \sim Q$$

is used to indicate that X has distribution Q ; that is, $P_X = Q$. The *cumulative distribution function* of X is defined by

$$F_X(x) = P(X \leq x) = P(\{e \in \mathcal{E} : X(e) \leq x\}) = P_X((-\infty, x]),$$

for $x \in \mathbb{R}$.

1.4 Null Sets

Let μ be a measure on $(\mathcal{X}, \mathcal{A})$. A set N is called *null* (or null with respect to μ) if

$$\mu(N) = 0.$$

If a statement holds for $x \in \mathcal{X} - N$ with N null, the statement is said to hold *almost everywhere* (a.e.) or a.e. μ . For instance, $f = 0$ a.e. μ if and only if $\mu(\{x \in \mathcal{X} : f(x) \neq 0\}) = 0$.

There is an alternative language for similar ideas when μ is a probability measure. Suppose some statement holds if and only if $x \in B$. Then the statement holds (a.e. μ) if and only if $\mu(B^c) = 0$ if and only if $\mu(B) = 1$. This can be expressed by saying “the statement holds *with probability one*.”

The values of a function on a null set cannot affect its integral. With this in mind, here are a few useful facts about integration that are fairly easy to appreciate:

1. If $f = 0$ (a.e. μ), then $\int f d\mu = 0$.
2. If $f \geq 0$ and $\int f d\mu = 0$, then $f = 0$ (a.e. μ).
3. If $f = g$ (a.e. μ), then $\int f d\mu = \int g d\mu$ whenever either one of the integrals exists.
4. If $\int 1_{(c,x)} f d\mu = 0$ for all $x > c$, then $f(x) = 0$ for a.e. $x > c$. The constant c here can be $-\infty$.

As a consequence of 2, if f and g are integrable and $f > g$, then $\int f d\mu > \int g d\mu$ (unless μ is identically zero).

1.5 Densities

Densities play a basic role in statistics. In many situations the most convenient way to specify the distribution of a random vector X is to give its density. Also, densities give likelihood functions used to compute Bayes estimators or maximum likelihood estimators. The density for a measure exists whenever it is absolutely continuous with respect to another measure according to the following definition.

Definition 1.9. *Let P and μ be measures on a σ -field $\mathcal{A}$ of $\mathcal{X}$. Then P is called absolutely continuous with respect to μ , written $P \ll \mu$, if $P(A) = 0$ whenever $\mu(A) = 0$.*

Theorem 1.10 (Radon–Nikodym). *If a finite measure P is absolutely continuous with respect to a σ -finite measure μ , then there exists a nonnegative measurable function f such that*

$$P(A) = \int_A f d\mu \stackrel{\text{def}}{=} \int f 1_A d\mu.$$

The function f in this theorem is called the Radon–Nikodym derivative of P with respect to μ , or the *density* of P with respect to μ , denoted

$$f = \frac{dP}{d\mu}.$$

By the third fact about integration and null sets in the previous section, the density f may not be unique, but if f_0 and f_1 are both densities, then $f_0 = f_1$ (a.e. μ). If $X \sim P_X$ and P_X is absolutely continuous with respect to μ with density $p = dP_X/d\mu$, it is convenient to say that X has *density* p with respect to μ .

Example 1.11. Absolutely Continuous Random Variables. If a random variable X has density p with respect to Lebesgue measure on $\mathbb{R}$, then X or its distribution P_X is called *absolutely continuous* with density p . Then, from the Radon–Nikodym theorem,

$$F_X(x) = P(X \leq x) = P_X((-\infty, x]) = \int_{-\infty}^x p(u) du.$$

Using the fundamental theorem of calculus, p can generally be found from the cumulative distribution function F_X by differentiation, $p(x) = F'_X(x)$.

Example 1.12. Discrete Random Variables. Let $\mathcal{X}_0$ be a countable subset of $\mathbb{R}$. The measure μ defined by

$$\mu(B) = \#(\mathcal{X}_0 \cap B)$$

for Borel sets B is (also) called *counting measure on $\mathcal{X}_0$* . As in Example 1.5,

$$\int f d\mu = \sum_{x \in \mathcal{X}_0} f(x).$$

Suppose X is a random variable and that

$$P(X \in \mathcal{X}_0) = P_X(\mathcal{X}_0) = 1.$$

Then X is called a *discrete random variable*. Suppose N is a null set for μ , so $\mu(N) = 0$. From the definition of μ , $\#(N \cap \mathcal{X}_0) = 0$, which means that $N \cap \mathcal{X}_0 = \emptyset$ and so $N \subset \mathcal{X}_0^c$. Then $P_X(N) = P(X \in N) \leq P(X \in \mathcal{X}_0^c) = 1 - P(X \in \mathcal{X}_0) = 0$. Thus N must also be a null set for P_X , and this shows that P_X is absolutely continuous with respect to μ . The density p of P_X with respect to μ satisfies

$$P(X \in A) = P_X(A) = \int_A p d\mu = \sum_{x \in \mathcal{X}_0} p(x) 1_A(x).$$

In particular, if $A = \{y\}$ with $y \in \mathcal{X}_0$, then $X \in A$ if and only if $X = y$, and so

$$P(X = y) = \sum_{x \in \mathcal{X}_0} p(x) 1_{\{y\}}(x) = p(y).$$

This density p is called the *mass function* for X . Note that because $\mathcal{X}_0^c$ is a null set, the density $p(y)$ can be defined arbitrarily when $y \notin \mathcal{X}_0$. The natural convention is to take $p(y) = 0$ for $y \notin \mathcal{X}_0$, for then $p(y) = P(Y = y)$ for all y .

1.6 Expectation

If X is a random variable on a probability space $(\mathcal{E}, \mathcal{B}, P)$, then the *expectation* or *expected value* of X is defined as

$$EX = \int X dP. \quad (1.4)$$

This formula is rarely used. Instead, if $X \sim P_X$ it can be shown that

$$EX = \int x dP_X(x).$$

Also, if $Y = f(X)$, then

$$EY = Ef(X) = \int f dP_X. \quad (1.5)$$

Integration against P_X in these two formulas is often accomplished using densities. If P_X has density p with respect to μ , then

$$\int f dP_X = \int fp d\mu. \quad (1.6)$$

This identity allows formal substitution of $p d\mu$ for dP_X , which makes the derivative notation $p = dP_X/d\mu$ seem natural. Together these results can all be viewed as change of variable results. Proofs of these results are based on the methods used to define integrals. It is easy to show that (1.5) and (1.6) hold when f is an indicator function. By linearity they must then hold for positive simple functions, and then a limiting argument shows that they hold for general measurable f , at least when the integrals exist. Specializing these results to absolutely continuous and discrete random variables we have the following important examples.

Example 1.13. If X is an absolutely continuous random variable with density p , then

$$EX = \int x dP_X(x) = \int xp(x) dx$$

and

$$Ef(X) = \int f(x)p(x) dx. \quad (1.7)$$

Example 1.14. If X is discrete with $P(X \in \mathcal{X}_0) = 1$ for a countable set $\mathcal{X}_0$, if μ is counting measure on $\mathcal{X}_0$, and if p is the mass function given by $p(x) = P(X = x)$, then

$$EX = \int x dP_X(x) = \int xp(x) d\mu(x) = \sum_{x \in \mathcal{X}_0} xp(x)$$

and

$$Ef(X) = \sum_{x \in \mathcal{X}_0} f(x)p(x). \quad (1.8)$$

Expectation is a linear operation. If X and Y are random variables and a and b are nonzero constants, then

$$E(aX + bY) = aEX + bEY, \quad (1.9)$$

provided EX and EY both exist and the right-hand side is not $\infty - \infty$. This follows easily from the definition of expectation (1.4), because integration is linear (1.3). Another important property of expectation is that if X and Y have finite expectations and $X < Y$ (a.e. P), then $EX < EY$. Also, using linearity and the second fact about integration in Section 1.4, if $X \leq Y$ (a.e. P) and both have finite expectations, $EX \leq EY$ with equality only if $X = Y$ (a.e. P).

The *variance* of a random variable X with finite expectation is defined as

$$\text{Var}(X) = E(X - EX)^2.$$

If X is absolutely continuous with density p , by (1.7)

$$\text{Var}(X) = \int (x - EX)^2 p(x) dx,$$

and if X is discrete with mass function p , by (1.8)

$$\text{Var}(X) = \sum_{x \in \mathcal{X}_0} (x - EX)^2 p(x).$$

Using (1.9),

$$\text{Var}(X) = E(X^2 - 2XEX + (EX)^2) = EX^2 - (EX)^2,$$

a result that is often convenient for explicit calculation.

The covariance between two random variables X and Y with finite expectations is defined as

$$\text{Cov}(X, Y) = E(X - EX)(Y - EY), \quad (1.10)$$

whenever the expectation exists. Note that $\text{Cov}(X, X) = \text{Var}(X)$. Using (1.9),

$$\begin{aligned} \text{Cov}(X, Y) &= E(XY - XEY - YEX + (EX)(EY)) \\ &= EXY - (EX)(EY). \end{aligned} \quad (1.11)$$

The covariance between two variables might be viewed as a measure of the linear association between the two variables. But because covariances are influenced by the measurement scale, a more natural measure is the *correlation*, defined using the covariance as

$$\text{Cor}(X, Y) = \frac{\text{Cov}(X, Y)}{[\text{Var}(X)\text{Var}(Y)]^{1/2}}.$$

Correlations always lie in $[-1, 1]$, with values ± 1 arising when there is a perfect linear relation between the two variables.²

1.7 Random Vectors

If $X_1, \dots, X_n$ are random variables, then the function $X : \mathcal{E} \rightarrow \mathbb{R}^n$ defined by

² This follows from the covariance inequality (4.11).

$$X(e) = \begin{pmatrix} X_1(e) \\ \vdots \\ X_n(e) \end{pmatrix}, \quad e \in \mathcal{E},$$

is called a *random vector*.³ Much of the notation and many of the results presented in this chapter for random variables extend naturally and directly to random vectors. For instance, the distribution P_X of X is defined by

$$P_X(B) = P(X \in B) \stackrel{\text{def}}{=} P(\{e \in \mathcal{E} : X(e) \in B\})$$

for Borel sets $B \in \mathbb{R}^n$, and notation $X \sim P_X$ means that X has distribution P_X . The random vector X or its distribution P_X is called *absolutely continuous* with density p if P_X is absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^n$. In this case

$$P(X \in B) = \int \cdots \int_B p(x) dx.$$

The random vector X is *discrete* if $P(X \in \mathcal{X}_0) = 1$ for some countable set $\mathcal{X}_0 \subset \mathbb{R}^n$. If $p(x) = P(X = x)$, then P_X has density p with respect to counting measure on $\mathcal{X}_0$ and

$$P(X \in B) = \sum_{x \in \mathcal{X}_0 \cap B} p(x).$$

The expectation of a random vector X is the vector of expectations,

$$EX = \begin{pmatrix} EX_1 \\ \vdots \\ EX_n \end{pmatrix}.$$

If $T : \mathbb{R}^n \rightarrow \mathbb{R}$ is a measurable function, then $T(X)$ is a random variable, and, as in (1.5),

$$ET(X) = \int T dP_X$$

whenever the expectation or integral exists. If P_X has a density p with respect to a dominating measure μ , this integral can be expressed as $\int T p d\mu$, which becomes

$$\sum_{x \in \mathcal{X}_0} T(x)p(x) \quad \text{or} \quad \int \cdots \int T(x)p(x) dx$$

in the discrete and absolutely continuous cases with μ counting or Lebesgue measure, respectively.

³ Equivalently, the vector-valued function X is measurable: $X^{-1}(B) \in \mathcal{E}$ for every Borel set $B \in \mathbb{R}^n$.

1.8 Covariance Matrices

A matrix W is called a *random matrix* if the entries W_{ij} are random variables. If W is a random matrix, then EW is the matrix of expectations of the entries,

$$(EW)_{ij} = EW_{ij}.$$

If v is a constant vector, A , B , and C are constant matrices, X is a random vector, and W is a random matrix, then

$$E[v + AX] = v + AEX \quad (1.12)$$

and

$$E[A + BWC] = A + B(EW)C. \quad (1.13)$$

These identities follow easily from basic properties of expectation because $(v + AX)_i = v_i + \sum_j A_{ij}X_j$ and $(A + BWC)_{ij} = A_{ij} + \sum_k \sum_l B_{ik}W_{kl}C_{lj}$.

The *covariance* of a random vector X is the matrix of covariances of the variables in X ; that is,

$$[\text{Cov}(X)]_{ij} = \text{Cov}(X_i, X_j).$$

If $\mu = EX$ and $(X - \mu)'$ denotes the transpose of $X - \mu$, a (random) row vector, then

$$\text{Cov}(X_i, X_j) = E(X_i - \mu_i)(X_j - \mu_j) = E[(X - \mu)(X - \mu)']_{ij},$$

and so

$$\text{Cov}(X) = E(X - \mu)(X - \mu)'. \quad (1.14)$$

Similarly, using (1.11) or (1.13),

$$\text{Cov}(X) = EXX' - \mu\mu'.$$

To find covariances after an affine transformation, because the transpose of a product of two matrices (or vectors) is the product of the transposed matrices in reverse order, using (1.14), (1.12), and (1.13), if v is a constant vector, A is a constant matrix, and X is a random vector, then

$$\begin{aligned} \text{Cov}(v + AX) &= E(v + AX - v - A\mu)(v + AX - v - A\mu)' \\ &= EA(X - \mu)(X - \mu)'A' \\ &= A[E(X - \mu)(X - \mu)']A' \\ &= A\text{Cov}(X)A'. \end{aligned} \quad (1.15)$$

1.9 Product Measures and Independence

Let $(\mathcal{X}, \mathcal{A}, \mu)$ and $(\mathcal{Y}, \mathcal{B}, \nu)$ be measure spaces. Then there exists a unique measure $\mu \times \nu$, called the *product measure*, on $(\mathcal{X} \times \mathcal{Y}, \mathcal{A} \vee \mathcal{B})$ such that

$$(\mu \times \nu)(A \times B) = \mu(A)\nu(B),$$

for all $A \in \mathcal{A}$ and all $B \in \mathcal{B}$. The σ -field $\mathcal{A} \vee \mathcal{B}$ is defined formally as the smallest σ -field containing all sets $A \times B$ with $A \in \mathcal{A}$ and $B \in \mathcal{B}$.

Example 1.15. If μ and ν are Lebesgue measures on $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively, then $\mu \times \nu$ is Lebesgue measure on $\mathbb{R}^{n+m}$.

Example 1.16. If μ and ν are counting measures on countable sets $\mathcal{X}_0$ and $\mathcal{Y}_0$, then $\mu \times \nu$ is counting measure on $\mathcal{X}_0 \times \mathcal{Y}_0$.

The following result shows that integration against the product measure $\mu \times \nu$ can be accomplished by iterated integration against μ and ν , in either order.

Theorem 1.17 (Fubini). *If $f \geq 0$, then*

$$\begin{aligned} \int f d(\mu \times \nu) &= \int \left[\int f(x, y) d\nu(y) \right] d\mu(x) \\ &= \int \left[\int f(x, y) d\mu(x) \right] d\nu(y). \end{aligned}$$

Dropping the restriction $f \geq 0$, if $\int |f| d(\mu \times \nu) < \infty$ then these equations hold.

Taking $f = 1_S$, this result gives a way to compute $(\mu \times \nu)(S)$ when S is not the Cartesian product of sets in $\mathcal{A}$ and $\mathcal{B}$.

Definition 1.18 (Independence). *Two random vectors, $X \in \mathbb{R}^n$ and $Y \in \mathbb{R}^m$ are independent if*

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B), \quad (1.16)$$

for all Borel sets A and B .

If $Z = \begin{pmatrix} X \\ Y \end{pmatrix}$, then $Z \in A \times B$ if and only if $X \in A$ and $Y \in B$, and (1.16) can be expressed in terms of the distributions of X , Y , and Z as

$$P_Z(A \times B) = P_X(A)P_Y(B).$$

This shows that the distribution of Z is the product measure,

$$P_Z = P_X \times P_Y.$$

The density of Z is also given by the product of the densities of X and Y . This can be shown using Fubini's theorem and (1.6) to change variables of integration. Specifically, suppose P_X has density p_X with respect to μ and P_Y has density p_Y with respect to ν . Then

$$\begin{aligned} P(Z \in S) &= \int 1_S d(P_X \times P_Y) \\ &= \int \left[\int 1_S(x, y) dP_X(x) \right] dP_Y(y) \\ &= \int \left[\int 1_S(x, y) p_X(x) d\mu(x) \right] p_Y(y) d\nu(y) \\ &= \int 1_S(x, y) p_X(x) p_Y(y) d(\mu \times \nu)(x, y). \end{aligned}$$

This shows that P_Z has density $p_X(x)p_Y(y)$ with respect to $\mu \times \nu$. In applications, μ and ν will generally be counting or Lebesgue measure. Note that the level of generality here covers mixed cases in which one of the random vectors is discrete and the other is absolutely continuous.

Whenever $Z = \begin{pmatrix} X \\ Y \end{pmatrix}$, P_Z is called the *joint distribution* of X and Y , and a density for P_Z is called the *joint density* of X and Y . So when X and Y are independent with densities p_X and p_Y , their joint density is $p_X(x)p_Y(y)$.

These ideas extend easily to collections of several random vectors. If Z is formed from random vectors $X_1, \dots, X_n$, then a density or distribution for Z is called a joint density or joint distribution, respectively, for $X_1, \dots, X_n$. The vectors $X_1, \dots, X_n$ are *independent* if

$$P(X_1 \in B_1, \dots, X_n \in B_n) = P(X_1 \in B_1) \times \cdots \times P(X_n \in B_n)$$

for any Borel sets $B_1, \dots, B_n$. Then $P_Z = P_{X_1} \times \cdots \times P_{X_n}$, where this product is the unique measure μ satisfying

$$\mu(B_1 \times \cdots \times B_n) = P_{X_1}(B_1) \times \cdots \times P_{X_n}(B_n).$$

The following proposition shows that functions of independent variables are independent.

Proposition 1.19. *If $X_1, \dots, X_n$ are independent random vectors, and if $f_1, \dots, f_n$ are measurable functions, then $f_1(X_1), \dots, f_n(X_n)$ are independent.*

If X_i has density p_{X_i} with respect to μ_i , $i = 1, \dots, n$, then $X_1, \dots, X_n$ have joint density p given by

$$p(x_1, \dots, x_n) = p_{X_1}(x_1) \times \cdots \times p_{X_n}(x_n)$$

with respect to $\mu = \mu_1 \times \cdots \times \mu_n$. If $X_1, \dots, X_n$ are independent, and they all have the same distribution, $X_i \sim Q$, $i = 1, \dots, n$, then $X_1, \dots, X_n$ are called *independent and identically distributed* (i.i.d.), and the collection of variables is called a *random sample* from Q .

1.10 Conditional Distributions

Suppose X and Y are random vectors. If X is observed and we learn that $X = x$, then P_Y should no longer be viewed as giving appropriate probabilities for Y . Rather, we should modify P_Y taking account of the new information that $X = x$. When X is discrete this can be accomplished using the standard formula for conditional probabilities of events: Let $p_X(x) = P(X = x)$, the mass function for X ; take $\mathcal{X}_0 = \{x : p_X(x) > 0\}$, the set of possible values for X ; and define

$$Q_x(B) = P(Y \in B | X = x) = \frac{P(Y \in B, X = x)}{P(X = x)} \quad (1.17)$$

for Borel sets B and $x \in \mathcal{X}_0$. Then for any $x \in \mathcal{X}_0$, it is easy to show that Q_x is a probability measure, called the *conditional distribution* for Y given $X = x$.

Formally, conditional probabilities should be *stochastic transition kernels*. These are defined as functions $Q : \mathcal{X} \times \mathcal{B} \rightarrow [0, 1]$ satisfying two properties. First, for $x \in \mathcal{X}$, $Q_x(\cdot)$ should be a probability measure on $\mathcal{B}$; and second, for any $B \in \mathcal{B}$, $Q_x(B)$ should be a measurable function of x .

For completeness, we should also define $Q_x(B)$ above when $x \notin \mathcal{X}_0$. How this is done does not really matter; taking Q_x to be some fixed probability measure for $x \notin \mathcal{X}_0$ would be one simple possibility.

Conditional distributions also exist when X is not discrete, but the definition is technical and is deferred to Chapter 6. However, the most important results in this section hold whether X is discrete or not. In particular, if X and Y are independent and X is discrete, by (1.17) Q_x equals P_Y , regardless of the value of $x \in \mathcal{X}_0$. This fact remains true in general and is the basis for a host of interesting and useful calculations.

Integration against a conditional distributions gives a conditional expectation. Specifically, the conditional expectation of $f(X, Y)$ given $X = x$ is defined as

$$E[f(X, Y) | X = x] = \int f(x, y) dQ_x(y). \quad (1.18)$$

Suppose X and Y are both discrete with Y taking values in a countable set $\mathcal{Y}_0$ and X taking values in $\mathcal{X}_0$ as defined above. Then $Z = \begin{pmatrix} X \\ Y \end{pmatrix}$ takes values in the countable set $\mathcal{X}_0 \times \mathcal{Y}_0$ and is discrete with mass function $p_Z(z) = P(Z = z) = P(X = x, Y = y)$, where $z = \begin{pmatrix} x \\ y \end{pmatrix}$. By (1.17), $Q_x(\mathcal{Y}_0) = 1$ and so Q_x is discrete with mass function q_x given by

$$q_x(y) = Q_x(\{y\}) = P(Y = y | X = x) = \frac{P(Y = y, X = x)}{P(X = x)} \quad (1.19)$$

for $x \in \mathcal{X}_0$. Then the conditional expectation in (1.18) can be calculated as a sum,

$$H(x) = E[f(X, Y) | X = x] = \sum_{y \in \mathcal{Y}_0} f(x, y) q_x(y).$$

For regularity, suppose $E|f(X, Y)| < \infty$. Noting from (1.19) that $P(X = x, Y = y) = q_x(y)p_X(x)$, the expectation of $f(X, Y)$ can be written as

$$\begin{aligned} Ef(X, Y) &= \sum_{\binom{x}{y} \in \mathcal{X}_0 \times \mathcal{Y}_0} f(x, y)P(X = x, Y = y) \\ &= \sum_{x \in \mathcal{X}_0} \sum_{y \in \mathcal{Y}_0} f(x, y)q_x(y)p_X(x) \\ &= \sum_{x \in \mathcal{X}_0} H(x)p_X(x) \\ &= EH(X). \end{aligned}$$

This is a fundamental result in conditioning, called the *law of total probability*, *the tower property*, or *smoothing*. In fact, smoothing identities are so basic that they form the basis for general definitions of conditional probability and expectation when X is not discrete. The random variable $H(X)$ obtained evaluating H from (1.18) at X is denoted

$$H(X) = E[f(X, Y) | X].$$

With this convenient notation the smoothing identity is just

$$Ef(X, Y) = EE[f(X, Y) | X].$$

In particular, when $f(X, Y) = Y$ this becomes

$$EY = EE(Y|X).$$

When $Y = 1_B$, the indicator of an event B , $EY = P(B)$ and this identity becomes

$$P(B) = EP(B|X),$$

where $P(B|X) \stackrel{\text{def}}{=} E(1_B|X)$. Finally, these identities also hold when the initial expectation or probability is conditional. Specifically,⁴

$$E(Y|X) = E[E(Y|X, W) | X] \tag{1.20}$$

and

$$P(B|X) = E[P(B|X, Y) | X].$$

⁴ See Problem 1.46.

1.11 Problems⁵

- *1. Prove (1.1). Hint: Define $A_1 = B_1$ and $A_n = B_n - B_{n-1}$ for $n \geq 2$. Show that the A_n are disjoint and use the countable additivity property of measure. Note: By definition, $\sum_{n=1}^{\infty} c_n = \lim_{N \rightarrow \infty} \sum_{n=1}^N c_n$.
2. For a set $B \subset \mathbb{N} = \{1, 2, \dots\}$, define

$$\mu(B) = \lim_{n \rightarrow \infty} \frac{\# [B \cap \{1, \dots, n\}]}{n},$$

when the limit exists, and let $\mathcal{A}$ denote the collection of all such sets.

- Find $\mu(E)$, $\mu(O)$, and $\mu(S)$, where $E = \{2, 4, \dots\}$, all even numbers, $O = \{1, 3, \dots\}$, all odd numbers, and $S = \{1, 4, 9, \dots\}$, all perfect squares.
 - If A and B are disjoint sets in $\mathcal{A}$, show that $\mu(A \cup B) = \mu(A) + \mu(B)$.
 - Is μ a measure? Explain your answer.
3. Suppose μ is a measure on the Borel sets of $(0, \infty)$ and that $\mu((x, 2x]) = \sqrt{x}$, for all $x > 0$. Find $\mu((0, 1])$.
4. Let $\mathcal{X} = \{1, 2, 3, 4\}$. Find the smallest σ -field $\mathcal{A}$ of subsets of $\mathcal{X}$ that contains the sets $\{1\}$ and $\{1, 2, 3\}$.
5. *Truncation.* Let μ be a measure on $(\mathcal{X}, \mathcal{A})$ and let A be a set in $\mathcal{A}$. Define ν on $\mathcal{A}$ by

$$\nu(B) = \mu(A \cap B), \quad B \in \mathcal{A}.$$

Show that ν is a measure on $(\mathcal{X}, \mathcal{A})$.

- Suppose $\mathcal{A}$ and $\mathcal{B}$ are σ -fields on the same sample space $\mathcal{X}$. Show that the intersection $\mathcal{A} \cap \mathcal{B}$ is also a σ -field on $\mathcal{X}$.
- Let $\mathcal{X}$ denote the rational numbers in $(0, 1)$, and let $\mathcal{A}$ be all subsets of $\mathcal{X}$, $\mathcal{A} = 2^{\mathcal{X}}$. Let μ be a real-valued function on $\mathcal{A}$ satisfying

$$\mu[(a, b) \cap \mathcal{X}] = b - a, \quad \text{for all } a < b, a \in \mathcal{X}, b \in \mathcal{X}.$$

Show that μ cannot be a measure.

- *8. Prove *Boole's inequality*: For any events $B_1, B_2, \dots$,

$$P\left(\bigcup_{i \geq 1} B_i\right) \leq \sum_{i \geq 1} P(B_i).$$

Hint: One approach would be to establish the result for finite collections by induction, then it extend to countable collections using (1.1). Another idea is to use Fubini's theorem, noting that if B is the union of the events, $1_B \leq \sum 1_{B_i}$.

⁵ Solutions to starred problems in each chapter are given at the back of the book.

9. *Cantor set.* The Cantor set can be defined recursively. Start with the closed unit interval $[0, 1]$ and form K_1 by removing the open middle third, so

$$K_1 = [0.1/3] \cup [2/3, 1].$$

Next, form K_2 by removing the two open middle thirds from the intervals in K_1 , so

$$K_2 = [0, 1/9] \cup [2/9, 1/3] \cup [2/3, 7/9] \cup [8/9, 1].$$

Continue removing middle thirds to form $K_2, K_3, \dots$. The Cantor set K is the limit or intersection of these sets,

$$K = \bigcap_{n=1}^{\infty} K_n.$$

Show that K is a Borel set and find its length or Lebesgue measure. Remark: K and $[0, 1]$ have the same cardinality.

- *10. Let μ and ν be measures on $(\mathcal{E}, \mathcal{B})$.

- a) Show that the sum η defined by $\eta(B) = \mu(B) + \nu(B)$ is also a measure.
b) If f is a nonnegative measurable function, show that

$$\int f d\eta = \int f d\mu + \int f d\nu.$$

Hint: First show that this result holds for nonnegative simple functions.

- *11. Suppose f is the simple function $1_{(1/2, \pi]} + 21_{(1, 2]}$, and let μ be a measure on $\mathbb{R}$ with $\mu\{(0, a^2]\} = a$, $a > 0$. Evaluate $\int f d\mu$.
*12. Suppose that $\mu\{(0, a)\} = a^2$ for $a > 0$ and that f is defined by

$$f(x) = \begin{cases} 0, & x \leq 0; \\ 1, & 0 < x < 2; \\ \pi, & 2 \leq x < 5; \\ 0, & x \geq 5. \end{cases}$$

Compute $\int f d\mu$.

- *13. Define the function f by

$$f(x) = \begin{cases} x, & 0 \leq x \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

Find simple functions $f_1 \leq f_2 \leq \dots$ increasing to f (i.e., $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ for all $x \in \mathbb{R}$). Let μ be Lebesgue measure on $\mathbb{R}$. Using our formal definition of an integral and the fact that $\mu((a, b]) = b - a$ whenever $b > a$ (this might be used to formally define Lebesgue measure), show that $\int f d\mu = 1/2$.

14. Suppose μ is a measure on $\mathbb{R}$ with $\mu([0, a]) = e^a$, $a \geq 0$. Evaluate $\int (1_{(0,2)} + 21_{[1,3]}) d\mu$.
15. Suppose μ is a measure on subsets of $\mathbb{N} = \{1, 2, \dots\}$ and that

$$\mu(\{n, n+1, \dots\}) = \frac{n}{2^n}, \quad n = 1, 2, \dots$$

Evaluate $\int x d\mu(x)$.

- *16. Define $F(a-) = \lim_{x \uparrow a} F(x)$. Then, if F is nondecreasing, $F(a-) = \lim_{n \rightarrow \infty} F(a - 1/n)$. Use (1.1) to show that if a random variable X has cumulative distribution function F_X ,

$$P(X < a) = F_X(a-).$$

Also, show that

$$P(X = a) = F_X(a) - F_X(a-).$$

- *17. Suppose X is a geometric random variable with mass function

$$p(x) = P(X = x) = \theta(1 - \theta)^x, \quad x = 0, 1, \dots,$$

where $\theta \in (0, 1)$ is a constant. Find the probability that X is even.

- *18. Let X be a function mapping $\mathcal{E}$ into $\mathbb{R}$. Recall that if B is a subset of $\mathbb{R}$, then $X^{-1}(B) = \{e \in \mathcal{E} : X(e) \in B\}$. Use this definition to prove that

$$X^{-1}(A \cap B) = X^{-1}(A) \cap X^{-1}(B),$$

$$X^{-1}(A \cup B) = X^{-1}(A) \cup X^{-1}(B),$$

and

$$X^{-1}\left(\bigcup_{i=0}^{\infty} A_i\right) = \bigcup_{i=0}^{\infty} X^{-1}(A_i).$$

- *19. Let P be a probability measure on $(\mathcal{E}, \mathcal{B})$, and let X be a random variable. Show that the distribution P_X of X defined by $P_X(B) = P(X \in B) = P(X^{-1}(B))$ is a measure (on the Borel sets of $\mathbb{R}$).
20. Suppose X is a Poisson random variable with mass function

$$p(x) = P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, \dots,$$

where $\lambda > 0$ is a constant. Find the probability that X is even.

- *21. Let X have a uniform distribution on $(0, 1)$; that is, X is absolutely continuous with density p defined by

$$p(x) = \begin{cases} 1, & x \in (0, 1); \\ 0, & \text{otherwise.} \end{cases}$$

Let Y_1 and Y_2 denote the first two digits of X when X is written as a binary decimal (so $Y_1 = 0$ if $X \in (0, 1/2)$ for instance). Find $P(Y_1 = i, Y_2 = j)$, $i = 0$ or 1 , $j = 0$ or 1 .

- *22. Let $\mathcal{E} = (0, 1)$, let $\mathcal{B}$ be the Borel subsets of $\mathcal{E}$, and let $P(A)$ be the length of A for $A \in \mathcal{B}$. (P would be called the *uniform probability measure* on $(0, 1)$.) Define the random variable X by

$$X(e) = \min\{e, 1/2\}.$$

Let μ be the sum of Lebesgue measure on $\mathbb{R}$ and counting measure on $\mathcal{X}_0 = \{1/2\}$. Show that the distribution P_X of X is absolutely continuous with respect to μ and find the density of P_X .

- *23. The standard normal distribution $N(0, 1)$ has density ϕ given by

$$\phi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}, \quad x \in \mathbb{R},$$

with respect to Lebesgue measure λ on $\mathbb{R}$. The corresponding cumulative distribution function is Φ , so

$$\Phi(x) = \int_{-\infty}^x \phi(z) dz$$

for $x \in \mathbb{R}$. Suppose that $X \sim N(0, 1)$ and that the random variable Y equals X when $|X| < 1$ and is 0 otherwise. Let P_Y denote the distribution of Y and let μ be counting measure on $\{0\}$. Find the density of P_Y with respect to $\lambda + \mu$.

- *24. Let μ be a σ -finite measure on a measurable space $(X, \mathcal{B})$. Show that μ is absolutely continuous with respect to some probability measure P . Hint: You can use the fact that if $\mu_1, \mu_2, \dots$ are probability measures and $c_1, c_2, \dots$ are nonnegative constants, then $\sum c_i \mu_i$ is a measure. (The proof for Problem 1.10 extends easily to this case.) The measures μ_i you will want to consider are truncations of μ to sets A_i covering X with $\mu(A_i) < \infty$, given by $\mu_i(B) = \mu(B \cap A_i)$. With the constants c_i chosen properly, $\sum c_i \mu_i$ will be a probability measure.

- *25. The monotone convergence theorem states that if $0 \leq f_1 \leq f_2 \leq \dots$ are measurable functions and $f = \lim f_n$, then $\int f d\mu = \lim \int f_n d\mu$. Use this result to prove the following assertions.

- a) Show that if $X \sim P_X$ is a random variable on $(\mathcal{E}, \mathcal{B}, P)$ and f is a nonnegative measurable function, then

$$\int f(X(e)) dP(e) = \int f(x) dP_X(x).$$

Hint: Try it first with f an indicator function. For the general case, let f_n be a sequence of simple functions increasing to f .

- b) Suppose that P_X has density p with respect to μ , and let f be a nonnegative measurable function. Show that

$$\int f dP_X = \int f p d\mu.$$

*26. *The gamma distribution.*

a) The gamma function is defined for $\alpha > 0$ by

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx.$$

Use integration by parts to show that $\Gamma(x+1) = x\Gamma(x)$. Show that $\Gamma(x+1) = x!$ for $x = 0, 1, \dots$.

b) Show that the function

$$p(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, & x > 0; \\ 0, & \text{otherwise,} \end{cases}$$

is a (Lebesgue) probability density when $\alpha > 0$ and $\beta > 0$. This density is called the gamma density with parameters α and β . The corresponding probability distribution is denoted $\Gamma(\alpha, \beta)$.

c) Show that if $X \sim \Gamma(\alpha, \beta)$, then $EX^r = \beta^r \Gamma(\alpha + r)/\Gamma(\alpha)$. Use this formula to find the mean and variance of X .

*27. Suppose X has a uniform distribution on $(0, 1)$. Find the mean and covariance matrix of the random vector $\begin{pmatrix} X \\ X^2 \end{pmatrix}$.

*28. If $X \sim N(0, 1)$, find the mean and covariance matrix of the random vector $\begin{pmatrix} X \\ I_{\{X > c\}} \end{pmatrix}$.

29. Let X be a random vector in $\mathbb{R}^n$ with $EX_i^2 < \infty$, $i = 1, \dots, n$, and let $A = EXX'$. Show that A is nonnegative definite: $v'Av \geq 0$ for all $v \in \mathbb{R}^n$.

30. Let W be absolutely continuous with density

$$p(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0; \\ 0, & \text{otherwise,} \end{cases}$$

where $\lambda > 0$ (the exponential density with failure rate λ), and define $X = \lfloor W \rfloor$ and $Y = W - X$. Here $\lfloor \cdot \rfloor$ is the *floor* or *greatest integer* function: $\lfloor x \rfloor$ is the greatest integer less than or equal to x .

a) Find $P(X = k)$, $k \geq 0$, the mass function for X .

b) Find $P(Y \leq y | X = k)$, $y \in (0, 1)$. What is the cumulative distribution function for Y ?

c) Find EY and $\text{Var}(Y)$.

d) Compute EW . Use linearity and your answer to (c) to find EX .

e) Find the covariance matrix for the random vector $\begin{pmatrix} Y \\ W \end{pmatrix}$.

31. Let X be an absolutely continuous random variable with density

$$p(x) = \begin{cases} 2x, & x \in (0, 1); \\ 0, & \text{otherwise.} \end{cases}$$

a) Find the mean and variance of X .

b) Find $E \sin(X)$.

c) Let $Y = I\{X > 1/2\}$. Find $\text{Cov}(X, Y)$.

*32. Suppose $E|X| < \infty$ and let

$$h(t) = \frac{1 - E \cos(tX)}{t^2}.$$

Use Fubini's theorem to find $\int_0^\infty h(t) dt$. Hint:

$$\int_0^\infty (1 - \cos(u))u^{-2} du = \frac{\pi}{2}.$$

*33. Suppose X is absolutely continuous with density $p_X(x) = xe^{-x}$, $x > 0$ and $p_X(x) = 0$, $x \leq 0$. Define $c_n = E(1 + X)^{-n}$. Use Fubini's theorem to evaluate $\sum_{n=1}^\infty c_n$.

34. Let Z have a standard normal distribution, introduced in Problem 1.23.

a) For $n = 1, 2, \dots$, show that

$$EZ^{2n} = (2n - 1)!! \stackrel{\text{def}}{=} (2n - 1) \times (2n - 3) \times \cdots \times 1.$$

Hint: Use an inductive argument based on an integration by parts identity or formulas for the gamma function.

b) Use the identity in (a) and Fubini's theorem to evaluate

$$\sum_{n=1}^\infty \frac{(2n - 1)!!}{3^n n!}.$$

35. Prove Proposition 1.19.

*36. Suppose X and Y are independent random variables, and let F_X and F_Y denote their cumulative distribution functions.

a) Use smoothing to show that the cumulative distribution function of $S = X + Y$ is

$$F_S(s) = P(X + Y \leq s) = EF_X(s - Y). \quad (1.21)$$

b) If X and Y are independent and Y is almost surely positive, use smoothing to show that the cumulative distribution function of $W = XY$ is $F_W(w) = EF_X(w/Y)$ for $w > 0$.

*37. Differentiating (1.21) with respect to s one can show that if X is absolutely continuous with density p_X , then $S = X + Y$ is absolutely continuous with density

$$p_S(s) = Ep_X(s - Y)$$

for $s \in \mathbb{R}$. Use this formula to show that if X and Y are independent with $X \sim \Gamma(\alpha, 1)$ and $Y \sim \Gamma(\beta, 1)$, then $X + Y \sim \Gamma(\alpha + \beta, 1)$.

*38. Let Q_λ denote the exponential distribution with failure rate λ , given in Problem 1.30. Let X be a discrete random variable taking values in $\{1, \dots, n\}$ with mass function

$$P(X = k) = \frac{2k}{n(n+1)}, \quad k = 1, \dots, n,$$

and assume that the conditional distribution of Y given $X = x$ is exponential with failure rate x ,

$$Y|X = x \sim Q_x.$$

- a) Find $E[Y|X]$.
- b) Use smoothing to compute EY .
- *39. Let X be a discrete random variable uniformly distributed on $\{1, \dots, n\}$, so $P(X = k) = 1/n$, $k = 1, \dots, n$, and assume that the conditional distribution of Y given $X = x$ is exponential with failure rate x .
 - a) For $y > 0$ find $P[Y > y|X]$.
 - b) Use smoothing to compute $P(Y > y)$.
 - c) Determine the density of Y .
- 40. Let X and Y be independent absolutely continuous random variables, X with density $p_X(x) = e^{-x}$, $x > 0$, $p_X(x) = 0$, $x < 0$, and Y uniformly distributed on $(0, 1)$. Let $V = X/(X + Y)$.
 - a) Find $P(V > c|Y = y)$ for $c \in (0, 1)$.
 - b) Use smoothing to compute $P(V > c)$.
 - c) What is the density of V ?
- 41. Suppose that X has the standard exponential distribution with density $p_X(x) = e^{-x}$, $x \geq 0$, $p_X(x) = 0$, $x < 0$; that Y has a (discrete) uniform distribution on $\{1, \dots, n\}$; and that X and Y are independent.
 - a) Find the joint density of X and Y . Use it to compute $P(X + Y > 3/2)$.
 - b) Find the covariance matrix for $Z = \begin{pmatrix} X \\ X + Y \end{pmatrix}$.
 - c) Find $E[\exp(XY/(1 + Y)) | X]$.
 - d) Use smoothing to compute $E \exp(XY/(1 + Y))$.
- 42. Two measures μ and ν on $(\mathcal{X}, \mathcal{A})$ are called (mutually) *singular* if $\mu(A) = \nu(A^c) = 0$ for some $A \in \mathcal{A}$. For instance, Lebesgue measure and counting measure on some countable subset $\mathcal{X}_0$ of $\mathbb{R}$ are singular (take $A = \mathcal{X}_0$). Let μ and ν be singular measures on the Borel sets of $\mathbb{R}$, and let Q_0 and Q_1 be probability measures absolutely continuous with respect to μ and ν , respectively, with densities

$$q_0 = \frac{dQ_0}{d\mu} \quad \text{and} \quad q_1 = \frac{dQ_1}{d\nu}.$$

Let X have a Bernoulli distribution with success probability p , and assume that

$$Y|X = 0 \sim Q_0 \quad \text{and} \quad Y|X = 1 \sim Q_1.$$

- a) Use the result in Problem 1.10 to show that Q_1 has density $q_1 1_A$ with respect to $\mu + \nu$, where $\mu(A) = \nu(A^c) = 0$.
- b) Use smoothing to derive a formula for $P(Y \in B)$ involving Q_0 , Q_1 , and p .

- c) Find a density for Y with respect to $\mu + \nu$.
43. Let X and Y be independent random variables with X uniformly distributed on $(0, 1)$ and Y uniformly distributed on $\{1, \dots, n\}$. Define $W = Ye^{XY}$.
- a) Find $E[W|Y = y]$.
- b) Use smoothing to compute EW .
44. The standard exponential distribution is absolutely continuous with density $p(x) = 1_{(0, \infty)}(x)e^{-x}$. Let X and Y be independent random variables, both from this distribution, and let $Z = X/Y$.
- a) For $z > 0$, find $P(Z \leq z|Y = y)$.
- b) Use smoothing and the result in part (a) to compute

$$P(Z \leq z), \quad z > 0.$$

- c) Find the covariance between Y and $I\{Z \leq z\}$.
45. Show that $E[f(X)Y|X] = f(X)E(Y|X)$.
46. If $E|Y| < \infty$ and f is a bounded function, then by smoothing,

$$E[f(X)Y] = E[f(X)E(Y|X)].$$

By (1.20) we should then have

$$E[f(X)Y] = E\left[f(X)E[E(Y|X, W)|X]\right].$$

Use a smoothing argument to verify that this equation holds, demonstrating that (1.20) works in this case.